

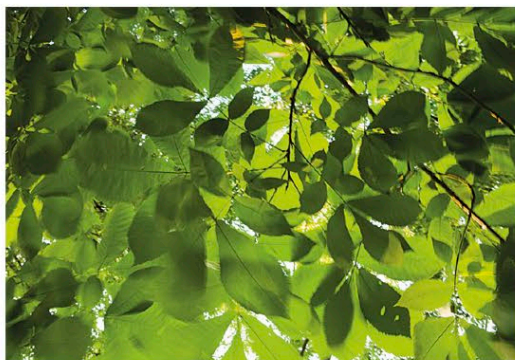
ORIGINS OF LIFE

The fossil record shows that life first appeared on Earth at least 3.7 billion years ago, and that all complex life has evolved from these first simple forms. Today, diverse life forms range from single-celled organisms to mammals with a complex anatomy.

WHAT IS LIFE?

Several features distinguish living organisms from inanimate, inorganic matter, although the existence of viruses blurs the boundary between the living and non-living. These features include an ability to take in and expend energy, to grow and change, to reproduce, to adapt to its environment, and – in more complex living organisms – to communicate.

The cell is the fundamental unit of life, capable of replicating itself and carrying out all living processes. Even the smallest independent organisms are made up of at least one cell, and almost every cell of every living organism has its own set of molecular instructions. Within each cell, the thread-like chromosomes carry hereditary information in the form of genes,



^ PHOTOSYNTHESIS

Plants use the pigment chlorophyll to capture light energy and convert water and carbon dioxide into sugars and oxygen. This benefits other life forms that feed on the plants and breathe in oxygen.

which are responsible for the particular characteristics of an organism. The set of instructions in a gene are mainly recorded in the form of a molecule called chromosomal deoxyribonucleic acid (DNA). The DNA of an organism carries information from one generation to another, allowing certain characteristics to be passed on from parent to offspring.

DIVISIONS OF LIFE

The huge array of life on Earth is divided into three domains or superkingdoms – Archaea, Bacteria, and Eukaryota – which encompass all life forms from plants and fungi to animals. Archaea and bacteria are primitive, single-celled organisms, and were probably the earliest form of life on Earth. The more advanced eukaryotes are distinguished from archaea and bacteria by having a cell nucleus, which contains the cell's genetic material, DNA.

Eukaryotes vary enormously in shape and size, ranging from single-celled organisms to complex, multi-celled plants and animals.

< GROWTH

A capacity to grow and repair is one of the key defining features of life. All organisms, from simple fungi to mammals, grow mostly by cell division but also increase in cell size.



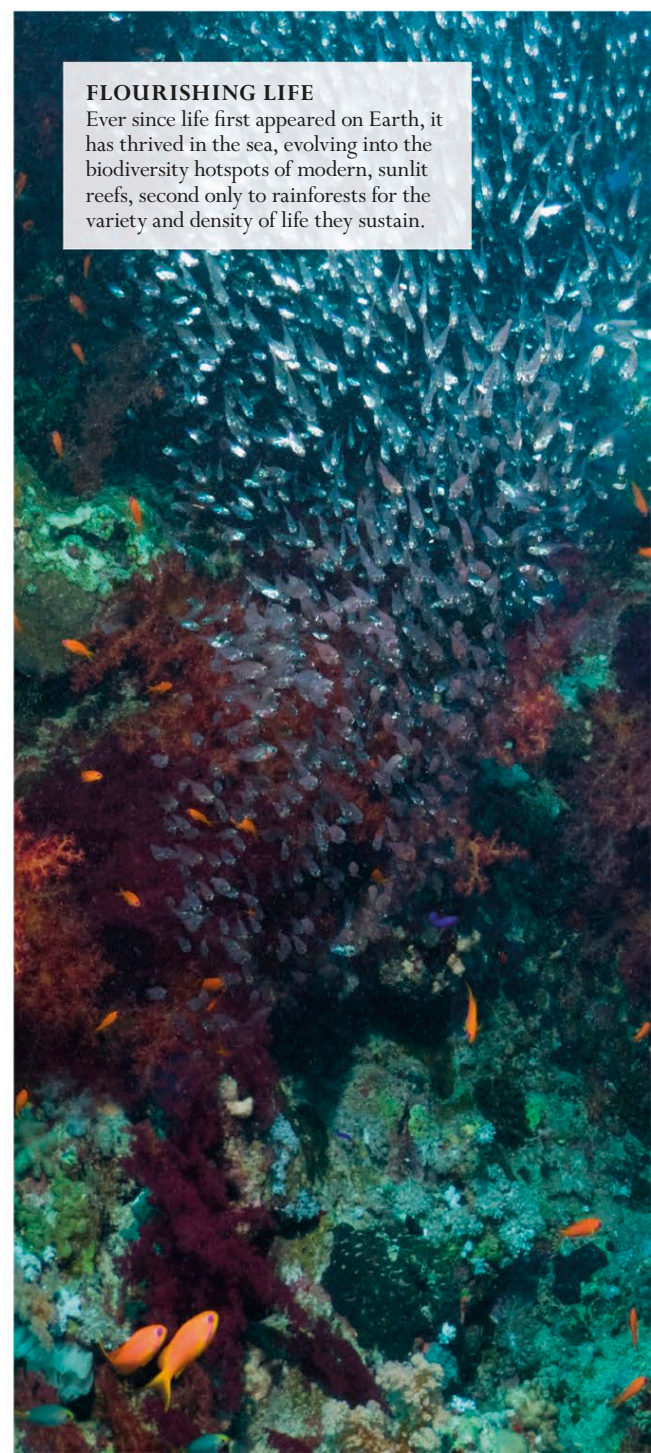
EARLY LIFE

The very earliest forms of life evolved in the seas, and evidence of this comes from two main sources: living primitive organisms and the fossil record. The most primitive life forms today are archaea and bacteria, which can survive even in extreme temperatures and acidic conditions. Such micro-organisms may be similar to those that first evolved in the extreme environments of early Earth.

Fossil evidence for early life is contentious. Chemical evidence of carbon derived from living organisms has been found in rocks from Western Greenland and dates back some 3.7 billion years ago. The most convincing of ancient records of life are layered, mound-shaped stromatolites (see right).

FLOURISHING LIFE

Ever since life first appeared on Earth, it has thrived in the sea, evolving into the biodiversity hotspots of modern, sunlit reefs, second only to rainforests for the variety and density of life they sustain.

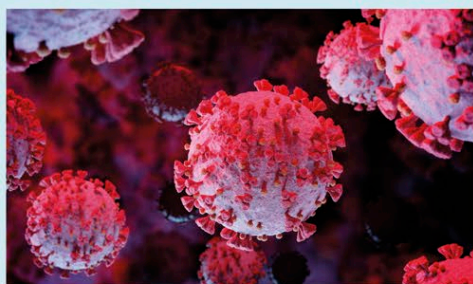


^ VITAL ENERGY

To sustain itself, life must obtain energy from the environment. Energy flows mostly from photosynthesis in plants – but also occasionally from chemosynthesis in bacteria – to the animals that feed upon them and are, in turn, eaten by other animals.

SCIENCE VIRUSES

Viruses are the most abundant biological entity on Earth, and they lie at the boundary between living and non-living. While they have features in common with living organisms – they are made of genetic material and are protected by a protein coat – viruses are parasitic and can only reproduce within the living cells of other organisms. They are tiny packets of chemicals that copy themselves, without truly being alive.



SARS-COV-2 VIRUS, THE CAUSE OF COVID-19